Nandiraju Gireesh

4th year Undergraduate Student
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Education

Birla Institute of Technology and Science – Pilani, Hyderabad Campus Telangana, India
B.E. in Electronics and Instrumentation Aug. 2017 – May 2021

Research Experience

Data Science Lab, BPHC **Advisor: Prof. Paresh Saxena**
Undergraduate Research Assistant starting Jan. 2021

- I will be working towards my Bachelor's Thesis in Reinforcement Learning on actor-critic algorithms.
- We are trying to improve the results of asynchronous advantage actor-critic algorithm (A3C) compared to previous methods such as A3C-GS, Averaged A3C, A3C-GAE, FFE, Adversary A3C.

AI-for-Healthcare Lab, UC San Diego **Advisor: Prof. Pengtao Xie**
Research Intern (Remote) Mar. 2020 – Aug. 2020

- We developed methods to generate view-consistent, high-fidelity, and high-resolution X-ray images from Radiology reports to facilitate radiology training of medical students. (Paper)
- Our method beats previous state-of-the-art methods like VanillaGAN, StackGAN, AttnGAN by 17.7% and 15.8% on OpenI & MIMIC-CXR datasets on Visual Consistency measure respectively.

Publications & Pre-prints

1. *XRayGAN: Consistency preserving Generation of X-Ray images from Radiology Reports*
   Xingyi Yang, **Nandiraju Gireesh**, Eric Xing, Pengtao Xie (Rejected AAAI 2021)

Skills

- **Programming Languages:** Python, Bash
- **Frameworks:** PyTorch, TensorFlow, Keras, fastai, OpenCV, ROS, OpenAI Gym, MuJoCo
- **Tools:** Kubernetes, Docker, GIT, Jupyter, Gazebo, Google Colab

Projects

Vehicle Detection and Tracking Project February 2020
(Object Detection, Computer Vision, Deep Learning, TensorFlow)

- Built a software pipeline which detects vehicles in a video stream and outputs an annotated video stream with a bounding box over vehicles detected.
- Performed a Histogram of Oriented Gradients (HOG) feature extraction on a labeled training set of images and trained them using a Linear Support Vector Machine Classifier (SVM).

- Combined my previous lane-finding algorithm to this project and produced a simultaneous lane-finding and vehicle detected annotated video as output.

Behavior Cloning February 2020

(Imitation Learning, Computer Vision, Deep Learning, TensorFlow)

- Trained a Deep Neural Network using TensorFlow that replicates human steering behavior while driving, thus being able to drive autonomously on a simulator with an accuracy of 98%.
- The model is partially based on NVIDIA's network architecture mentioned in their research paper "End to End Learning for Self-Driving Cars".
- This network takes as input the frame of the frontal camera (say a roof mounted camera) and predicts the steering direction at each instant.

Traffic Sign Recognition Classifier January 2020

(Image Recognition, Computer Vision, Deep Learning, TensorFlow)

- Built and trained a Deep Neural Network which decodes traffic signs from natural images by using German Traffic Sign Dataset.
- The model is based on LeNet Architecture, which is a Convolutional Neural Network designed to recognize visual patterns directly from pixel images with minimal pre-processing.
- Implemented the model using TensorFlow and Scikit-learn's Pipeline framework, which showed an accuracy of 96% on Test Dataset with 500 epochs on CPU.

Finding Lane Lines on the road for Autonomous Vehicles January 2020

(Image Processing, Computer Vision, Python)

- Built a pipeline that identifies lane lines taking road images from video as input and returns an annotated video as output.
- Applied Canny Edge Detection, Region of Interest Selection and Hough Transform Line Detection techniques and processed the video clips to find lane lines in them.
- Used Python and OpenCV to build the pipeline and the whole model was built in Jupyter Notebook.

Relevant Courses (Online)

- CS 285 - Deep Reinforcement Learning – Prof. Sergey Levine (Current)
- Coursera – Reinforcement Learning – University of Alberta (Current)
- Practical Deep Learning for Coders – fast.ai
- Deep Learning from the Foundations – fast.ai

Scholarships

Prime Minister's Scholarship Scheme (May 2018 - Present)